# Nickel Purification of Nanobodies

Reagents:

- SET Lysis Buffer (200 mM Tris pH 8, 500 mM Sucrose, 0.5 mM EDTA)
- Cold $H_2O$
- Benzonase Nuclease
- 3 M $MgCl_2$
- 5 M NaCl
- Wash Buffer 1: 20 mM Hepes pH 7.5, **500 mM NaCl**, 20 mM imidazole
    - The high salt wash removed non-specific binders from the nickel column
- Wash Buffer 2: 20 mM Hepes pH 7.5, 100 mM NaCl, 20 mM imizaole
- Elution Buffer 20 mM Hepes pH 7.5, 100 mM NaCl, 200 mM imidazole
- 10 kDa cutoff concentrator (Amicon)
- Size Exclusion / Dialysis Buffer (20 mM Hepes pH 7.5, 150 mM NaCl, 10% Glycerol)
- Ni-NTA Resin

## Protocol

Expression:

1. Transform pET26b-Nb plasmid (Kan resistance) into T7 express lysY competent BL21 E.coli (NEB C3010)
2. Pick one colony and grow an overnight culture in LB with 50µg/mL Kan, 2% Glucose, and 1mM $MgCl_2$
3. Inoculate 1L of TB in 2.8L baffled flask containing 0.1% Glucose, 1mM MgCl2 and 50µg/mL Kan with 10mL of overnight culture
4. Grow until an OD600 of 0.6-0.8 at 37°C, inoculate with 1mM IPTG and grow overnight at 17°C
5. Pellet cells at 4000xRPM and freeze pellet in liquid N2 and store at -80°C

Purification:

1. Resuspend cell pellet in room temperature SET buffer. Use 50ml buffer per L of culture volume
2. Stir for 45 minutes
3. Add 2x volume of cold water. Water should be added quickly to induce osmotic shock. Add 5 mM $MgCl_2$, 1 µL benzonase
4. Stir 1 hr at room temperature
5. Spin at 20,000 x g for 30 min (For large culture volume, you can use 1 L centrifuge bottles, and spin at 8000g for longer)
6. To the supernatant add NaCl to 100 mM, imidazole to 20 mM
7. Stir at room temperature for 15 min
8. Filter supernatant through glass microfiber filter
9. Apply supernatant to Ni-NTA resin equilibrated with wash buffer
    - For gravity purification use ~3-5 mL resin / L culture volume, load at a moderate speed, to increase yield collect flow through run over the column again to ensure full binding

10. Wash with 10 CV of wash buffer 1
11. Wash with 10 CV of wash buffer 2
12. Elute with 10-20 CV of elution buffer
13. Wash resin with 5-10 more CV of elution buffer to clean if you are planning to use resin again
14. Concentrate protein containing fractions in 10 kDa cutoff concentrator. If nanobody precipitates clear by spinning at 15-20,000 x g for 5 minutes at 4°C or filter with 0.22 um filter
15. Some nanobodies that precipitate can be rescued by the addition of extra salt (300-500 mM)
16. Run over superdex S75 or dialyze overnight to buffer exchange.
17. Check purify by SDS-PAGE
18. Occasionally, a second band may be present that is slightly larger than the expected molecular weight of the nanobody. This may be the nanobody with a retained PelB sequence. This can be separated by ion exchange, but will need to be optimized for individual nanobodies (e.g. MonoS column in 20 mM Hepes pH 7.5 with slow gradient between 450-520 mM NaCl for nanobody with a pI of 9.4.)
19. Note: Nanobodies for crystallization can be treated with carboxypeptidases A and B to remove C-terminal His-Tag